



DMISRS M&E Report 2021

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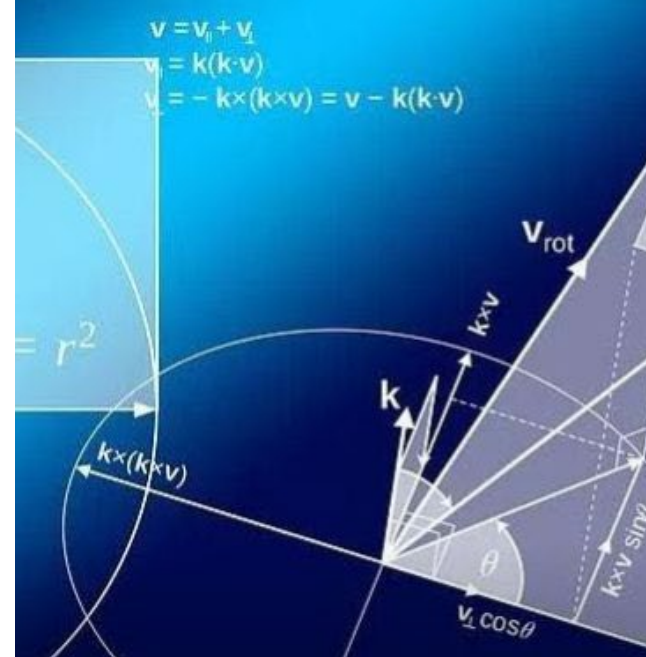


Table of Contents

Introduction	3
The Diagnostic Mathematics Information for Student Retention and Success (DMISRS) Project.....	3
Project Outcomes.....	4
DMISRS Project Activities	7
HEI Support Survey	7
DMISRS Symposium	7
DMISRS Qualitative Study	9
Going Forward.....	10
Conclusion & Recommendations.....	10

Introduction

This report summarises the project activities, outputs and participant outcomes of the 2021 Diagnostic Mathematics Information for Student Retention and Success (DMISRS) Project. In particular, it describes the mid-year symposium as well as a qualitative study that was undertaken at the end of the year to gauge project outcomes. The project continues to be well-received by partners, and suggestions made by partners for project improvement have been reflected on by the DMISRS team and incorporated into future planning.

What follows is an introduction to the project and a presentation of key DMISRS activities, as well as the associated outputs and participant outcomes per activity.

The Diagnostic Mathematics Information for Student Retention and Success (DMISRS) Project

The DMISRS Project was developed to address the lack of alignment between students' literacy practices and mathematical proficiency, and the academic practices of their chosen disciplines. This gap results in significant attrition, which continues to widen inequality, and impacts the economy and future development (about 40% of all registered students have dropped out of their studies by the end of regulation time, and only 27% obtain their degrees in the regulation time [CHE, 2013]). The project proposal sought to contribute towards the national imperative to address the problem of high failure and drop-out rates, in STEM programmes, specifically in first year Mathematics courses.

The DMISRS Project focuses specifically on support for students' mathematical needs in higher education and in STEM programmes in particular, given the national imperative. The project aims to analyse the NBT Mathematics scores and Higher Education Management Information System (HEMIS) data sets in order to establish how students' performance in the NBTs correlate with graduation and drop-out rates. It also aims to research the effect of curriculum-integrated support initiatives for Mathematics in the first year of study and initiate the development of blended learning resources aligned with this research. The ability to achieve these aims hinged on the expansion of National Benchmark Tests Project (NBTP) Mathematics (MAT) testing to facilitate the inclusion of an expanded range of South African Universities who would not typically have NBTP MAT data for their registered cohorts, and hence optimise and expand the data available.

Broadly, the project involves the following:

1. Individual areas of need for students in identified institutions are determined through sub-domain analysis of NBT Mathematics results in Year 1;
2. Determining the support for Mathematics that has been available in the past (for example, a hot seat, or tutorial classes) in an at-risk Mathematics course in identified institutions are determined;
3. The inclusion of additional support (for example open educational resource material or other online topic-specific resources mediated by a blended learning specialist) based on what seems to have worked at other institutions will be considered;
4. At the end of Year 1, the results will be scrutinised to determine whether they are better than /same as /worse than before (even though this will be a new student group it should be indicative of the impact of the initiative).

Project Outcomes

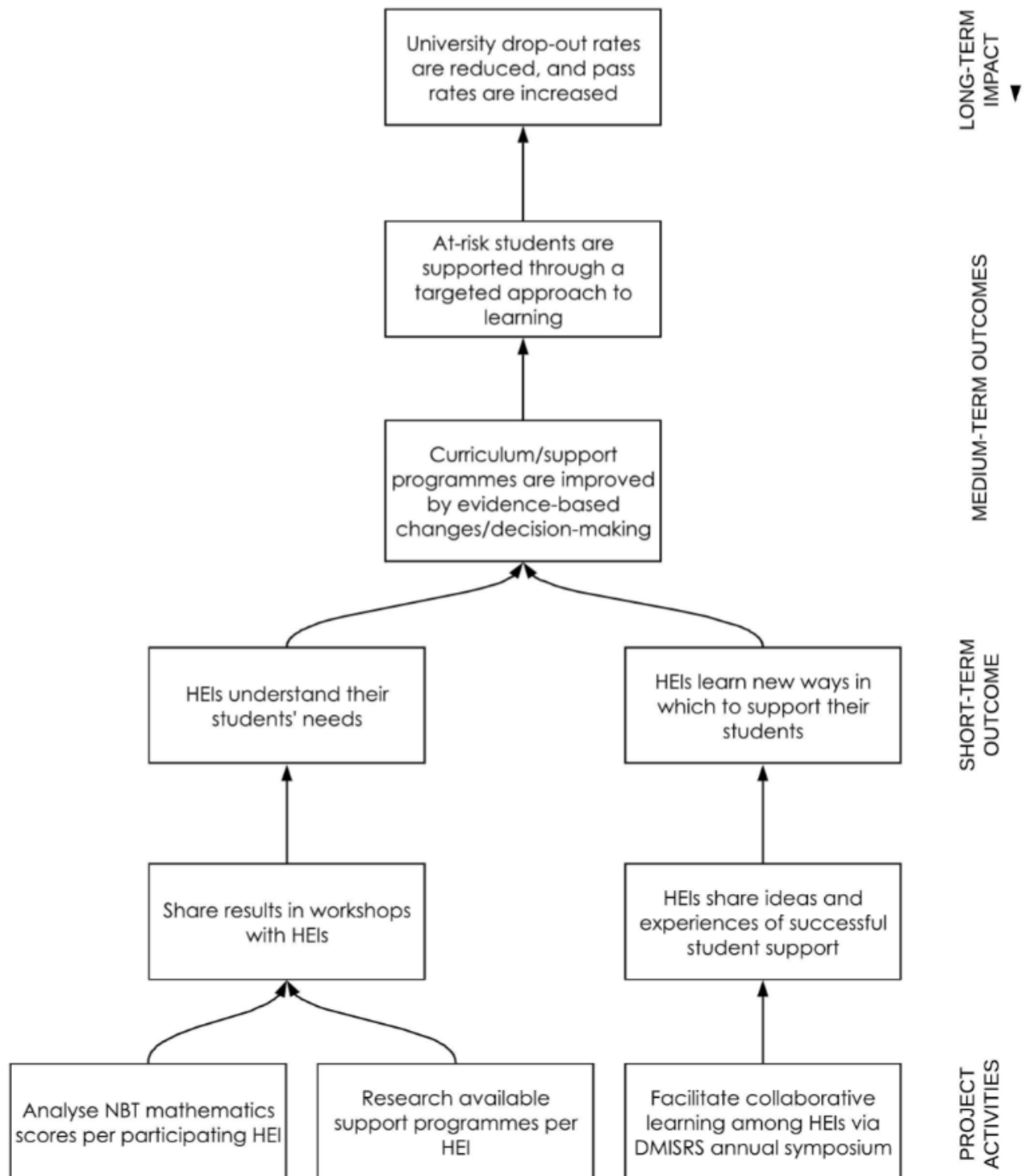
The project's outcomes are best served by a Theory of Change that describes the sequence in which these outcomes are expected to be achieved. A Theory of Change further helps to ensure project activities are aligned with the goals of the project, and guides the outputs and participant outcomes that should be measured and monitored. A Theory of Change for the DMISRS project was developed by external consultants in collaboration with the DMISRS team in February, 2019. It is visualized on the following page, and described below.

The Theory of Change begins with the project's primary activities: (1) analyse first-year students' NBT mathematics scores at each of the participating HEIs in relation to their first-year performance, and institutional drop-out and graduation rates; and (2) investigate what curriculum-integrated support programmes are available to these students at each HEI, and what are best practices in this regard. The results of these two investigations are then shared with participating HEIs at annual Teaching and Learning workshops.

As a result of these activities, and direct participation in the annual workshops, HEIs develop an understanding of their students' performance and, consequently, the gaps in their

support programmes that fail to address these needs. Using this knowledge, HEIs improve their current student support programmes by using the evidence presented to them in the workshops.

At-risk students are thus better supported by their faculties' revised targeted approach to learning, and, as a result, it is expected that drop-out rates will decrease, and pass/graduation rates will increase.



DMISRS Project Activities

HEI Support Survey

Every year, the DMISRS team endeavors to survey project participants regarding the student support mechanisms that are available to students. In 2020, sixteen respondents completed the survey from 13 institutions. In 2021, the survey was distributed in September but only four respondents completed it, prohibiting any meaningful analysis. This reflects an ongoing trend of survey fatigue among project participants; there has been a steady decline in survey feedback during the COVID-19 pandemic.

As such, it is recommended that feedback mechanisms are built into project activities, as opposed to post-activity exercises. During the symposium and institutional workshops, the DMISRS team can invite participants to engage in an in-person feedback session. Prompts for facilitating the discussion can be as follows:

- What was your biggest takeaway from this session?
- What was interesting or useful to you?
- What was missing?
- What could be improved?

DMISRS Symposium

The DMISRS symposium was held online on 26 July 2021. The symposium consisted of eight presentations and two keynote addresses centered around the theme for the year: *first year mathematics: from crises mode to the new 'normal'*

Themes that emerged across presentations and discussions include:

- The importance of placing students at the center and aligning teaching strategies with students' academic needs and circumstances;
- The critical role of formative diagnostic data in lieu of the above;
- The challenges involved in online student assessment; and
- The myriad of technological innovations and practices that can effectively support student engagement and the online learning environment.

All presentations and discussions were summarised by the project's M&E team and compiled into a report (available upon request).

At the end of the symposium, participants were asked to complete a short feedback survey. The purpose of the survey was to gauge partners' perceptions of the usefulness of the symposium, whether they intended to apply what they had learned, and how to improve future symposia. The survey results are used by the DMISRS team to measure the impact of their work, and to inform continuous improvement of DMISRS offerings. Twenty symposium attendees opted to provide this feedback.

The surveys were analysed and written up into a report by the M&E team. Key takeaways include:

- Regarding the effectiveness of the novel online format, the majority of survey respondents rated it as *very good* (30%) or *excellent* (55%).
- All participants said that they found a discussion/presentation particularly interesting or useful. The most frequently cited presentation was the keynote address by Dr Pragashni Padayachee (*Teaching and learning Mathematics during Covid 19: Reflections on a new 'normal'*).
- Two commonly cited takeaways from the symposium included: (1) the need to embrace technologies and innovative solutions that are available for teaching and learning; and (2) the value of the academic community in terms of sharing common experiences and challenges (depicted in the word cloud in Figure 1).
- The majority of participants (75%) reported that they intend to incorporate what they had learned at the symposium into their future planning.
- Overall, the symposium appeared to be well-received and a valuable offering from the CEA team. Some respondents closed off the survey by thanking the team for an interesting and effective event.

perspectives, experiences and common challenges have been a source of growth and comfort to project participants, especially as higher education has moved online.

"I think that the symposia as well as the workshops were useful to me in the sense that they gave me a better picture of what others experience; sometimes there are differences and sometimes there are similarities...It's useful to hear other people's voices and experiences."

While measuring the impact of the DMISRS Project is difficult in such fluid and emergent conditions, and as many events have had to be cancelled or changed due to the pandemic, a definite sense of learning and positive growth were noted among the interviewees. Furthermore, every interviewee had a compliment to bestow on the DMISRS team and the project in general, indicating widespread appreciation for the project and the lengths that the team are willing to go to in order to support the members.

"It's been good, supportive...it's motivational that the project exists. It's always done with a very good sense of care, good follow-up, calls for papers, it's well organised. It's a nice comforting space."

Going Forward: M&E

Monitoring and evaluation of the DMISRS project and associated activities will continue in 2022. This includes: analysing and reporting on participant feedback; reporting on Teaching and Learning Workshops/webinars; reporting on the DMISRS Symposium; and compiling an annual M&E report.

Conclusion & Recommendations

While tertiary education still grapples with a 'new normal', and DMISRS participants are understandably disrupted or occupied, the project still saw great levels of attendance at its

annual symposium, as well as positive sentiments from attendees. Furthermore, the qualitative study highlighted the significant connecting role played by the project, and the positive impact that it has had on participants.

Going forward, the M&E team has the following recommendations, based off of feedback from the project participants:

For the symposium -

- In-person engagement for the 2022 symposium, or greater efforts to stimulate interaction between the presenters and audience (e.g., breakaway groups, Q&A).
- Invite a more diverse audience to the symposium, especially those who hold the power to implement changes within the first-year courses or their management.

Events –

- Plan additional institutional workshops, if feasible, as these prove to be very useful, well-received and requested by participants.
- Facilitate an online webinar early on in the year on the most recent diagnostic results, and how the cohort compares to others (especially those pre-COVID). This can incorporate ideas around how best to plan for the current cohort in terms of academic gaps and online support.

Feedback

- At all events, plan a short session in which participants can give their immediate feedback about the event.

Funding

- Distribute information to participants regarding the funding that is available, and what it can be used for.

Community

- Consider creating an online resource hub in which DMISRS participants can network and access past reports, symposia presentations, etc., as well as share resources and upcoming events.